

Exercises 2.3

Unless otherwise specified, assume that all matrices in these exercises are $n \times n$. Determine which of the matrices in Exercises 1-10 are invertible. Use as few calculations as possible. Justify your answers.

3. $\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix} \xrightarrow[\times 1/5]{\text{Row 1} = \text{Row 1}} \begin{bmatrix} 1 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ + 3 \times \text{Row 1}}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ - 8 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 5 & -1 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & -7 & 0 \\ 0 & 5 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 2} = \text{Row 2} \\ \times (-1/7)}]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 5 & -1 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ - 5 \times \text{Row 2}}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix} \xrightarrow[\times (-1)]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

By The Invertible Matrix Theorem and $b \Rightarrow a$, the matrix $\begin{bmatrix} 5 & 0 & 0 \\ -3 & -7 & 0 \\ 8 & 5 & -1 \end{bmatrix}$ is invertible.

5. $\begin{bmatrix} 0 & 3 & -5 \\ 1 & 0 & 2 \\ -4 & -9 & 7 \end{bmatrix} \xrightarrow[\text{Row 2 are interchanged}]{\text{Row 1 and Row 2 are interchanged}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ -4 & -9 & 7 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ + 4 \times \text{Row 1}}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ 0 & -9 & 15 \end{bmatrix} \xrightarrow[\substack{\text{Row 3} = \text{Row 3} \\ \text{Row 3} + 3 \times \text{Row 2}}]{\text{Row 3} = \text{Row 3}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & 0 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 3 & -5 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow[\times 1/3]{\text{Row 2} = \text{Row 2}} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -5/3 \\ 0 & 0 & 0 \end{bmatrix}$